



# UNIVERSITY OF THE PUNJAB

## BSCS 3<sup>rd</sup> Semester

Roll No . . . . .

### CC-112-L Programming Fundamentals – 2024 (PRACTICAL)

Subject: Programming Fundamentals (CC-112L) Time: 3Hrs. Marks: 100

#### Group I

**Note: There are three parts in the practical. All parts are compulsory.**

#### **PART-I (MAX. MARKS: 30)**

##### **Question 01**

**[20 Marks]**

Write a function that accepts two parameters, an array of integers and its size, the function returns the average of all elements which are divisible by 3 in the array.

##### **Question 02 Viva Voce**

**[10 Marks]**

#### **PART-II (MAX. MARKS: 30)**

##### **Question 1**

**[20 Marks]**

Write a function named `isPrime`, which takes an integer as an argument and returns `true` if the argument is a prime number, or `False` otherwise. [only write the definition of this function]

##### **Question 02 Viva Voce**

**[10 Marks]**

#### **PART-III (MAX. MARKS: 40)**

##### **Question 01**

**[25 Marks]**

Assume yourself as a part of windows live Hotmail. Your job is to determine the password length of the user entered passwords.

A Password:

- Is said to be short if it contains less than 6 characters.
- Is perfect if it contains characters greater than 6 (max. length is 20 characters).

You are to design a function that takes a parameter string (By Reference) and determines its length according to the above-mentioned criteria.

##### **Question #02 Viva Voce**

**[15 Marks]**